

Polycythemia / MPN — Essential thrombocythemia



1. Platelets >1,000,000

WHAT YOU SEE

>1,000,000 mound of platelets in dock

WHAT IT ENCODES

Essential thrombocythemia (ET) is a clonal myeloproliferative neoplasm of megakaryocytes producing sustained thrombocytosis. The WHO diagnostic threshold is a platelet count $\geq 450 \times 10^9/L$ (450K), but counts often run >1,000K. Diagnosis (WHO 2016) requires all 4 major criteria OR first 3 major + 1 minor: (1) platelets $\geq 450K$, (2) marrow with proliferation of enlarged mature hyperlobulated megakaryocytes without significant granulocytic/erythroid increase, (3) not meeting criteria for PV/PMF/CML/MDS, (4) JAK2/CALR/MPL mutation; minor = a clonal marker or no reactive cause. Mnemonic: '450 is the gate, a million is the picture.'

BOARD TRAP

WRONG answer: 'platelets >100K is enough to diagnose ET.' The discriminator is the WHO threshold of $\geq 450K$ AND demonstration of clonality with exclusion of reactive causes — mild thrombocytosis (450–600K) is far more often reactive (iron deficiency, inflammation, infection, post-splenectomy) than ET.

2. JAK2 ~64% → PV risk

WHAT YOU SEE

REVELATION BOARD 64% JAK2+ ET = PV risk (women)

WHAT IT ENCODES

JAK2 V617F is the most common ET driver, found in ~50–65% of cases. JAK2-mutated ET tends to behave more like 'forme fruste PV' — higher hemoglobin, lower platelet counts for a given disease burden, more thrombosis, and a risk of progressing to overt PV. It is enriched in women. Mnemonic: 'JAK2-ET leans toward the red side (PV).'

BOARD TRAP

The trap is assuming JAK2 positivity proves ET — WRONG, since JAK2 V617F is shared across PV (~95%) and PMF (~50%). Discriminator: JAK2+ with isolated thrombocytosis, normal/low Hct, and characteristic large mature megakaryocytes on marrow defines ET; the same mutation with high RCM is PV, and with marrow fibrosis is PMF.

3. CALR → PMF risk (men)

WHAT YOU SEE

CALR = PMF RISK TYPE 1 (MEN)

WHAT IT ENCODES

CALR (calreticulin) mutations occur in ~20–25% of ET (the second most common driver) and are mutually exclusive with JAK2. CALR-mutated ET is associated with younger men, higher platelet counts, LOWER thrombosis risk, but a greater tendency toward fibrotic (PMF) transformation, especially type 1 CALR. CALR-mutant disease often carries a better overall prognosis than JAK2-mutant. Mnemonic: 'CALR — Counts high, cLots Low, fibrosis risk.'

BOARD TRAP

WRONG to think CALR-positive ET is higher-risk for thrombosis because the platelets are so high. Discriminator: despite the highest platelet counts, CALR+ patients have LOWER thrombotic risk than JAK2+ patients; the platelet number alone does not drive thrombosis risk — driver mutation, age, and prior thrombosis do (IPSET-thrombosis score).

4. MPL & exclusion of others

WHAT YOU SEE

MPL SF3B1, EXCLUSION Dx, ISOLATED-NOT ET

WHAT IT ENCODES

MPL (thrombopoietin receptor) mutations account for ~3–5% of ET. ET is fundamentally a DIAGNOSIS OF EXCLUSION: rule out CML (BCR-ABL1 fusion must be absent), MDS with ring sideroblasts and thrombocytosis (MDS/MPN-RS-T, often SF3B1-mutated), PV, PMF, and reactive thrombocytosis. About 10–15% of ET are 'triple-negative' (no JAK2/CALR/MPL). Mnemonic: 'EXCLUDE before you conclude — BCR-ABL must be negative.'

BOARD TRAP

The most important board trap: a high platelet count with basophilia, splenomegaly, and a left-shifted leukocytosis is CML, NOT ET — and the WRONG answer is to start hydroxyurea for 'ET.' Discriminator: ALWAYS test BCR-ABL1; a positive Philadelphia chromosome reclassifies the patient to CML and mandates a tyrosine kinase inhibitor (imatinib).

5. Thrombosis & bleeding

WHAT YOU SEE

vWF strands around platelet mound, bleeding pool below

WHAT IT ENCODES

ET produces a paradox of BOTH thrombosis and bleeding. Thrombosis (arterial > venous, including unusual sites like splanchnic/Budd-Chiari) is the leading complication. But at extreme counts (>1,000–1,500K) the huge platelet mass adsorbs and clears high-molecular-weight von Willebrand multimers, causing ACQUIRED von Willebrand disease and mucocutaneous bleeding. Erythromelalgia (burning red painful hands/feet) is the classic microvascular symptom, exquisitely aspirin-responsive. Mnemonic: 'too many platelets clot AND bleed.'

BOARD TRAP

WRONG answer: 'ET causes only thrombosis, never bleeding.' Discriminator: at extreme thrombocytosis, screen for acquired vWD (ristocetin cofactor activity) BEFORE giving aspirin — adding aspirin to a patient with acquired vWD can precipitate major bleeding. The bleeding-thrombosis paradox is the high-yield catch.

6. Aspirin

WHAT YOU SEE

ASPIRIN bottle bottom, red X for extreme counts

WHAT IT ENCODES

Low-dose aspirin (81 mg) relieves vasomotor symptoms (erythromelalgia, headache, visual disturbance) and reduces thrombosis; it is standard in low-risk patients with vasomotor symptoms and in all higher-risk patients. Mnemonic: 'aspirin melts erythromelalgia.'

BOARD TRAP

Giving aspirin reflexively to every ET patient is WRONG when platelets are extreme (>1,000–1,500K) and acquired vWD is present (the red X). Discriminator: check ristocetin cofactor activity first — if acquired vWD is documented, hold aspirin and lower the platelet count with cytoreduction before reintroducing antiplatelet therapy.

7. Hydroxyurea (high-risk)

WHAT YOU SEE

PEG/HU/ANAGRELIDE bottles right

WHAT IT ENCODES

Cytoreduction is indicated for HIGH-RISK ET (age >60 with JAK2, OR any prior thrombosis). Hydroxyurea is first-line: it lowers platelets, reduces thrombosis, is well tolerated; side effects include myelosuppression, megaloblastic changes, leg ulcers, and skin cancers. Pegylated interferon-alfa is preferred in young patients and pregnancy (non-teratogenic, non-leukemogenic). Risk stratification uses IPSET-thrombosis (age, prior thrombosis, JAK2, CV risk factors). Mnemonic: 'high-risk = Hydroxyurea; young/pregnant = Interferon.'

BOARD TRAP

WRONG to cytoreduce a low-risk young ET patient just because the platelet count is alarmingly high. Discriminator: treatment is driven by THROMBOTIC RISK (age >60, prior thrombosis, JAK2 status), NOT by the absolute platelet number — low-risk asymptomatic patients are managed with observation ± aspirin and CV risk-factor control.

8. Anagrelide

WHAT YOU SEE

ANAGRELIDE bottle

WHAT IT ENCODES

Anagrelide is a phosphodiesterase-3 inhibitor that selectively reduces platelet production by impairing megakaryocyte maturation. It is a SECOND-LINE cytoreductive agent. Side effects: headache, palpitations/tachycardia, fluid retention, and high-output cardiac effects. The PT-1 trial showed anagrelide + aspirin was inferior to hydroxyurea + aspirin, with more arterial thrombosis, bleeding, and progression to myelofibrosis. Mnemonic: 'Anagrelide = platelet-selective but second-string.'

BOARD TRAP

Choosing anagrelide first-line over hydroxyurea in high-risk ET is the WRONG answer. Discriminator: hydroxyurea + aspirin is first-line; anagrelide is reserved for hydroxyurea intolerance/resistance because PT-1 data show more thrombosis and faster myelofibrotic transformation with anagrelide.

9. Normalizing count ≠ prevent thrombosis

WHAT YOU SEE

NORMALIZING COUNT ≠ PREVENT THROMBOSIS banner

WHAT IT ENCODES

Driving the platelet count fully to the normal range does NOT abolish thrombotic risk in ET — thrombosis is driven by clonal platelet dysfunction, leukocyte activation, JAK2 status, age, and conventional CV risk factors, not by the count alone. The therapeutic target in high-risk patients is roughly <400K, but aspirin and CV risk-factor control remain essential. Mnemonic: 'the number is not the whole risk.'

BOARD TRAP

WRONG assumption: 'if I normalize the platelet count, the patient cannot clot.' Discriminator: thrombotic risk persists at normal counts; ongoing aspirin, smoking cessation, BP/lipid/diabetes control, and avoiding estrogen are required — fixating on the platelet number alone misses the dominant risk drivers.

10. Survival ~ general population

WHAT YOU SEE

SURVIVAL BAR ET ≈ general population, <60 = no greater than controls

WHAT IT ENCODES

ET has the BEST prognosis of the classic MPNs — median survival approaches that of the age-matched general population, especially in patients under 60. Leukemic/myelofibrotic transformation is low (~1–5% to AML, slightly higher to post-ET MF over decades). Mnemonic: 'ET = the most indolent MPN, lives for decades.'

BOARD TRAP

WRONG answer: 'ET has the worst MPN survival.' Discriminator: ET is the most indolent classic MPN with near-normal life expectancy; primary myelofibrosis carries the worst prognosis. Confusing ET's benign course with PMF's aggressive one is a common tested error.

11. Marrow: clustered megakaryocytes

WHAT YOU SEE

BM BIOPSY megakaryocyte hyperplasia, splenomegaly suggests another MPN

WHAT IT ENCODES

ET marrow shows proliferation of LARGE, MATURE, hyperlobulated (staghorn) megakaryocytes in loose clusters, with normal or near-normal erythroid and granulocytic lines and NO significant reticulin fibrosis. Significant reticulin fibrosis or dense/atypical megakaryocyte clusters point to early/pre-fibrotic PMF rather than true ET. Massive splenomegaly is unusual in ET and should prompt reconsideration of another MPN. Mnemonic: 'big mature staghorn megs, no fibrosis = ET.'

BOARD TRAP

WRONG to call a thrombocytotic patient with marrow reticulin fibrosis and dense atypical megakaryocyte clustering 'ET.' Discriminator: those features define prefibrotic primary myelofibrosis (pre-PMF), which has worse prognosis and higher transformation risk; the marrow morphology is what separates ET from pre-PMF.

12. Demographics 1-2/100K, female

WHAT YOU SEE

DEMOGRAPHICS 1-2/100K FEMALE

WHAT IT ENCODES

ET incidence is ~1–2.5 per 100,000/year with a female predominance and a bimodal pattern — a peak around 50–60 years and a smaller peak in young women of childbearing age. Because young women are affected, pregnancy management (aspirin, sometimes interferon, increased miscarriage/thrombosis risk) is a tested scenario. Mnemonic: 'ET = the MPN of younger women.'

BOARD TRAP

WRONG to assume a pregnant woman with ET should receive hydroxyurea for cytoreduction. Discriminator: hydroxyurea and anagrelide are teratogenic/contraindicated in pregnancy; the correct agents are low-dose aspirin and, if cytoreduction is needed, interferon-alfa (which is pregnancy-safe).

13. Pseudohyperkalemia artifact

WHAT YOU SEE

serum K+ FALSELY HIGH, NORMAL plasma, ARTIFACT

WHAT IT ENCODES

Extreme thrombocytosis causes SPURIOUS (pseudo) hyperkalemia: as the clotted serum sample forms, platelets lyse and release intracellular potassium, falsely elevating the serum K+. The PLASMA potassium (heparinized, unclotted tube) is normal — that is the diagnostic clue. Similar pseudo-elevations can affect serum phosphate. Mnemonic: 'platelets pop in the clot tube and spill K+.'

BOARD TRAP

WRONG answer: treat the high potassium as true hyperkalemia with insulin/dextrose, calcium, and kayexalate. Discriminator: an ET patient with high serum K+ but a NORMAL ECG should have a plasma K+ checked — a normal plasma value confirms pseudohyperkalemia, an in-vitro artifact requiring no treatment. Treating it can cause iatrogenic hypokalemia.

14. Iron stores / exclude IDA

WHAT YOU SEE

RETICULIN, ABSENT IRON STORES flagged NEEDS EXPLANATION

WHAT IT ENCODES

Absent marrow iron stores / low ferritin in a thrombocytotic patient demands explanation. Two key scenarios: (1) iron-deficiency itself causes REACTIVE thrombocytosis, which must be excluded before diagnosing ET; (2) coexisting iron deficiency can MASK polycythemia vera — the iron-limited erythropoiesis keeps the Hct deceptively normal while thrombocytosis is the only visible clue. Repleting iron in masked PV can unmask a dangerous Hct rise. Mnemonic: 'low iron — reactive thrombocytosis OR hidden PV.'

BOARD TRAP

WRONG to diagnose ET in an iron-deficient patient without addressing the iron status. Discriminator: correct/evaluate iron deficiency first — reactive thrombocytosis resolves with iron repletion (not ET), and in a JAK2+ patient, iron repletion may unmask PV (Hct surges), changing both the diagnosis and the management.
